

Manifold optimization is the problem of minimizing a function whose domain is not flat Euclidean space but a smooth Riemannian manifold — a curved geometric structure where the notion of straight-line movement must be replaced by geodesics and where gradients live not in ambient space but in the tangent space at each point. Besides minimizing a function, this approach allows us, after modifying the cost function, to explore the manifold - meaning to sample new points on the manifold.

Classical gradient-based methods like steepest descent extend to this setting via Riemannian gradients and retractions, but they require the cost function to be differentiable for the gradient to be calculated at each iteration step, which is often impractical and very time consuming.

Šnášel, Kong & Das address this in "From constraints fusion to manifold optimization: A new directional transport manifold metaheuristic algorithm" by proposing mDTMA, the first metaheuristic optimizer designed natively for Riemannian manifolds. The algorithm is derivative-free and population based: at each iteration a tournament selection identifies promising parents, a Simulated Binary Crossover (SBX) and Polynomial Mutation (PM) operator — borrowed from the PlatEMO evolutionary framework — generates candidate displacement vectors in ambient Euclidean space, and each vector is then projected onto the tangent space at the corresponding particle's current position and mapped back onto the manifold via a retraction, guaranteeing geometric validity throughout. A decaying inertia weight $w = w_0 + 0.1 * (1 - t/T)$ interpolates between broad global exploration early in the run and local refinement near convergence, mirroring the role of the inertia weight in Particle Swarm Optimization.

After modifying the algorithm, we are able to change mDTMA algorithm's function to a different task: instead of converging to a single optimum, the algorithm is driven to collect a maximally spread dataset of points covering the manifold as uniformly as possible. This is achieved by replacing the objective with a repulsion potential — the negative minimum distance from any candidate point to all previously accepted points — turning tournament selection into a diversity-maximizing pressure. A permanent archive ensures no region is revisited, a minimum-spacing rejection filter prevents near-duplicate entries, and parent fitness is re-evaluated against the growing archive before each selection step so that the pressure to explore new territory compounds over time.

The result is a principled sampler whose output constitutes training data for manifold learning algorithms such as (UMAP, Isomap, etc.), which require dense and geometrically uniform coverage of the underlying low-dimensional structure to generalize faithfully.